

TRIXINE POWDER

Ingredient(s):

Each g contains :

Sulfadimethoxine 400mg(potency)
Trimethoprim 80mg(potency)

Preservative(s):

Each g contains :

Sodium Benzoate 1.0mg

Pharmacology (Summary of Pharmacodynamics and Pharmacokinetics):

Sulfadimethoxine is a sulfonamide with long-acting and broad spectrum of activity which also anticoccidial properties. It is active against Gram-positive and Gram-negative and some protozoa such as coccidia.

Trimethoprim belongs to the family of diaminopyrimidines. It is active against most streptococci and Gram-negative bacteria.

In combination, these two active substances are synergistic. Sulfadimethoxine is potentiated by a diaminopyrimidine, trimethoprim. The combination of these two active substances allows a sequential blockade of the biosynthesis of folic acid. Both substances act sequentially on the route of synthesis of tetrahydrofolic acid: the sulfonylurea in inhibiting the incorporation of para-aminobenzoic acid in folic acid, trimethoprim by specifically inhibiting microbial dihydrofolate reductase. The theoretical spectrum of activity extends to both Gram-positive (*Staphylococcus*, *Listeria*, etc.) and Gram-negative (*Escherichia coli*, *Salmonella*, *Proteus*, *Enterobacter*, *Bordetella*, etc).

Sulfadimethoxine is considered a sulphonylurea with a delay longer persistence in plasma. Its protein binding is important. Distribution is good in most tissues and organs. Trimethoprim is rapidly absorbed after oral administration. It is widely distributed in the body. The two active substances are partially metabolised in the liver. Elimination is primarily renal.

Indication(s):

Rabbits and poultry:

- 1) Infections caused by or associated with organisms sensitive to sulphadimethoxine and trimethoprim.
- 2) Curative and preventive treatment of respiratory and digestive infections.
- 3) Curative and preventive treatment of digestive coccidiosis.

Target Species

Rabbits and poultry.

Dosage and Administration:

0.5–1g per liter of drinking water, or per 10kg of body weight, during 5 consecutive days. Orally via drinking water.

Contraindications:

Except under special circumstances, this medication should not be used when the following medical problems exist:

- 1) Blood dyscrasias: Slight to moderate reduction in hematopoietic activity has been reported with long-term high dosing of potentiated sulfonamides.
- 2) Hypersensitivity to diaminopyrimidines or sulfonamides: Animals that have had a previous reaction may be much more likely to react on subsequent administration.

Precaution(s) / Warning(s):

- 1) Trimethoprim/sulfa should not be used in dogs or horses showing marked liver parenchymal damage, blood dyscrasias, or in those with a history of sulfonamide sensitivity.
- 2) An idiosyncratic sulfonamide toxicosis can occur in any breed of dog, but the reaction has been reported more frequently in the Doberman Pinscher than in other breeds. This specific type of drug reaction includes blood dyscrasias, nonseptic polyarthritis, and skin rash.
- 3) Risk-benefit should be considered when the following medical problems exist:
 - a. Hepatic function impairment: Delayed biotransformation may increase the risk of adverse effects.
 - b. Renal function impairment: Delayed elimination could cause accumulation of sulfonamide and metabolites, increasing the risk of adverse effects.
 - c. Urolithiasis: Sulfonamides can crystallize in the renal system under certain conditions.

Interaction with Other Medicaments

- 1) Trimethoprim/sulfa may prolong the clotting times in patients receiving coumarin (warfarin) anticoagulants.
- 2) Sulfonamides may displace other highly bound drugs., such as methotrexate, phenylbutazone, thiazide diuretics, salicylates, probenecid and phenytoin.
- 3) Antacids may decrease the bioavailability of sulfonamides if administered concurrently.
- 4) Trimethoprim may decrease the therapeutic effect of cyclosporine (systemic) and increase the risk of nephrotoxicity developing.
- 5) A trimethoprim and sulfonamide combination administered to a detomidine-anesthetized horse can lead to arrhythmias, hypotension, and death; it is suspected that the antimicrobial potentiates the cardiac changes reported with detomidine.

Pregnancy and Lactation

Sulfonamides and diaminopyrimidines cross the placenta in pregnant animals and some teratogenic effects have been seen with very high doses given to pregnant mice and rats.

Side Effect(s) / Adverse Reaction(s):

- 1) Adverse effects noted in dogs: keratoconjunctivitis sicca (which may be irreversible), acute neutrophilic hepatitis with icterus, vomiting, anorexia, diarrhea, fever, hemolytic anemia, urticaria, polyarthritis, facial swelling, polydipsia, polyuria and cholestasis.
- 2) Potentiated sulfonamides may cause hypothyroidism in the dog, particularly with extended therapy.
- 3) Acute hypersensitivity reactions manifesting as Type I (anaphylaxis) or Type III reaction (serum sickness) can also be seen. Hypersensitivity reactions appear to be more common in large breed dogs; Doberman Pinschers may possibly be more susceptible to this effect than other breeds.
- 4) Adverse effects noted in cats may include anorexia, leukopenias and anemias.
- 5) Sulfonamides (or their metabolites) can precipitate in the urine, particularly when given at high dosages for prolonged periods. Acidic urine or highly concentrated urine may also contribute to increased risk of crystalluria, hematuria and renal tubule obstruction.
- 6) Oral therapy has resulted in diarrhea development in some horses. Previous administration of potentiated sulfas have been implicated in increasing the mortality rate of equine diarrhea.

Environmental Property

None has been mentioned in animal.

Symptoms and Treatment for Overdosage, and Antidote(s):

- 1) Manifestations of an acute overdosage can include symptoms of GI distress (nausea, vomiting, diarrhea), CNS toxicity (depression, headache, and confusion), facial swelling, bone marrow depression and increases in serum aminotransferases.
- 2) Long-term (8 weeks) therapy at recommended doses resulted in elevated serum cholesterol, thyroid and liver weights, mild follicular thyroid hyperplasia and enlarged basophilic cells in the pituitary.
- 3) In dogs, doses greater than 80mg/kg resulted in slight tremors and increased motor activity in some dogs. Higher doses may result in depression, anorexia or seizures.
- 4) Recommended treatment of overdose consists of the following:
 - a. Discontinuing medication.
 - b. Administering intravenous diazepam or other acute antiseizure medication, as needed.
 - c. Providing fluid replacement therapy as required.

Shelf-Life:

4 years.

Storage Condition(s):

Store at temperature below 30°C.

Used within 24 hours after reconstitution.

Product Description and Packing(s):

White to pale-yellow powder and without other foreign matter.

Al. Pouch of 500g

For food producing animals product:

Withdrawal Periods

Poultry: 6 days

Rabbits: 8 days

Eggs: As no withdrawal period for the eggs has been established, do not use in layers when eggs are delivered to human consumption (4 weeks prior and during laying)

Maximum Residual Limit (MRL):

Sulfonamides : All food producing species: Muscle, fat, liver, kidney and Cattle: Milk
100µg/kg

Trimethoprim : All food producing species except horses: Muscle, fat, liver, kidney, milk
50µg/kg
Horses: Muscle, fat, liver, kidney 100µg/kg



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